NOTRE DAME UNIVERSITY BANGLADESH

Computer Graphics Assignment-02

Course Code: CSE-4204
Course Title: Computer Graphics Lab
Lab Task Topic: Midpoint & Bresenhum Circle Algorithm

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Assignment No 2

Question: Suppose, center $(-2,4)$ & Radius 5, Compute all the points of the circle using Midpoint Circle Algorithm.

Solution: Given, $(x_0, y_0) = (0, 5)$ $\quad r = 5$

$\therefore P = 1 - 5 = -4$

Calculation table for 1st octant:

No	(x_k, y_k)	P_k	(x_{k+1}, y_{k+1})	P_{k+1}
0	$(0, 5)$	-4	$(1, 5)$	$P_1 = -4 + 2\cdot 0 + 3 = -1$
1	$(1, 5)$	-1	$(2, 5)$	$P_2 = -1 + 2\cdot 1 + 3 = 4$
2	$(2, 5)$	4	$(3, 4)$	$P_3 = 4 + 2(2-5) + 5 = 3$
3	$(3, 4)$	3	$(4, 3)$	$P_4 = 3 + 2(3-4) + 5 = 6$

$\therefore$ All the point of 1st Octant are,

(x_k, y_k)	$x_p = (x_c + x_k)$, $y_p = (y_c + y_k)$
$(0, 5)$	$(-2, 9)$
$(1, 5)$	$(-1, 9)$
$(2, 5)$	$(0, 9)$
$(3, 4)$	$(1, 8)$

1st octant $\rightarrow$ $(-2,9)$, $(-1,9)$, $(0,9)$, $(1,8)$.

Table for 1st Quadrant:

(x_p, y_p)	(y_p, x_p)
(-2, 9)	(9, -2)
(-1, 9)	(9, -1)
(0, 9)	(9, 0)
(1, 8)	(8, 1)

So, All the points of 1st Quadrant are,

(-2,9), (-1,9), (0,9), (1,8), (9,-2), (9,-1), (9,0), (8,1)

Table for all point:

(+,+)	(-,+)	(-,-)	(+,-)
(-2,9)	(2,9)	(2,-9)	(-2,-9)
(-1,9)	(1,9)	(1,-9)	(-1,-9)
(0,9)	(0,9)	(0,-9)	(0,-9)
(1,8)	(-1,8)	(-1,-8)	(1,-8)
(9,-2)	(-9,-2)	(-9,2)	(9,2)
(9,-1)	(-9,-1)	(-9,1)	(9,1)
(9,0)	(-9,0)	(-9,0)	(9,0)
(8,1)	(-8,1)	(-8,-1)	(8,-1)

So, all the points of the circle are: (-2,9), (2,9), (2,-9), (-2,-9), (-1,9), (1,9), (1,-9), (-1,-9), (0,9), (0,-9), (1,8), (-1,8), (-1,-8), (1,-8), (9,-2), (-9,-2), (-9,2), (9,2), (9,-1), (-9,-1), (-9,1), (9,1), (9,0), (-9,0), (8,1), (-8,1), (-8,-1), (8,-1).

(Ans)

Question: Suppose, center $(-2,4)$ & radius 5. Compute all the points of the circle using Bresenhum circle Algorithm.

$x_c = -2$

$y_c = 4$

Solution: Let, $x_0 = 0$; $y_0 = r = 5$

$$P_0 = 3 - 2r = 3 - 2 \cdot 5 = -7$$

No.	(x_k, y_k)	P_k	(x_{k+1}, y_{k+1})	P_{k+1}	(x_p, y_p)
0	(0, 5)	-7	(1, 5)	$P_1 = -7+4\cdot1+6 = 3$	(-2, 9)
1	(1, 5)	3	(2, 4)	$P_2 = 3+4(2-4)+10 = 5$	(-1, 9)
2	(2, 4)	5	(3, 3)	$P_3 = 5+4(3-3)+10 = 15$	(0, 8)
3	(3, 3)	15	(4, 2)	$P_4 = 15+4(4-2)+10 = 33$	(1, 7)
4	(4, 2)	33	(5, 1)	$P_5 = 33+4(5-1)+10 = 59$	(2, 6)
5	(5, 1)	59	(6, 0)	$P_6 = 59+4(6-0)+10 = 93$	(3, 5)
6	(6, 0)	93	(7, -1)	$P_7 = 93+4(7+1)+10 = 135$	(4, 4)

All the points of the 1st octant are,

(-2, 9), (-1, 9), (0, 8), (1, 7), (2, 6), (3, 5), (4, 4).

Table for 1st Quadrant:

(x_p, y_p)	(y_p, x_p)
(-2, 9)	(9, -2)
(-1, 9)	(9, -1)
(0, 8)	(8, 0)
(1, 7)	(7, 1)
(2, 6)	(6, 2)
(3, 5)	(5, 3)
(4, 4)	(4, 4)

So all the points of 1st Quadrant are: (-2,9), (-1,9), (0,8), (1,7), (2,6), (3,5), (4,4), (9,-2), (9,-1), (8,0), (7,1), (6,2), (5,3)

Table for all points:

(+,+)	(-,+)	(-,-)	(+,-)
(-2,9)	(2,9)	(2,-9)	(-2,-9)
(-1,9)	(1,9)	(1,-9)	(-1,-9)
(0,8)	(0,8)	(0,-8)	(0,-8)
(1,7)	(-1,7)	(-1,-7)	(1,-7)
(2,6)	(-2,6)	(-2,-6)	(2,-6)
(3,5)	(-3,5)	(-3,-5)	(3,-5)
(4,4)	(-4,4)	(-4,-4)	(4,-4)
(9,-2)	(-9,-2)	(-9,2)	(9,2)
(9,-1)	(-9,-1)	(-9,1)	(9,1)
(8,0)	(-8,0)	(-8,0)	(8,0)
(7,1)	(-7,1)	(-7,-1)	(7,-1)
(6,2)	(-6,2)	(-6,-2)	(6,-2)
(5,3)	(-5,3)	(-5,-3)	(5,-3)

So, all the point of circle are: (-2,9), (2,9), (2,-9), (-2,-9), (-1,9), (1,9), (1,-9), (-1,-9), (0,8), (0,-8), (1,7), (-1,7), (-1,-7), (1,-7), (2,6), (-2,6), (-2,-6), (2,-6), (3,5), (-3,5), (-3,-5), (3,-5), (4,4), (-4,4), (-4,-4), (4,-4), (9,-2), (-9,-2), (-9,2) (9,2), (9,-1), (-9,-1), (-9,1), (9,1), (8,0), (-8,0), (7,1), (-7,1), (-7,-1), (7,-1), (6,2), (-6,2), (-6,-2), (6,-2), (5,3), (-5,3), (-5,-3), (5-3)

(Ans)